

Decent Coaching Classes
Praneet Science Objective Questions
Class VIII (NCERT) – Pressure, Winds, Storms and Cyclones

A. Multiple Choice Questions (Choose the correct option)

1. Pressure is defined as the force acting per unit _____.
(a) volume (b) area (c) time (d) length
2. The SI unit of pressure is:
(a) Newton (b) Joule (c) Watt (d) Pascal
3. One pascal is equal to:
(a) 1 N/m (b) 1 N m (c) 1 N/m² (d) 1 kg/m²
4. Which formula correctly gives pressure?
(a) $P = F \times A$ (b) $P = A/F$ (c) $P = F + A$ (d) $P = F/A$
5. If the force remains the same and the area is doubled, the pressure becomes:
(a) double (b) half (c) unchanged (d) four times
6. If the area remains the same and the force increases, the pressure:
(a) increases (b) decreases (c) remains the same (d) becomes zero
7. Pressure is inversely proportional to:
(a) force (b) area (c) weight (d) mass
8. A school bag with broad straps is more comfortable because the straps:
(a) reduce the weight of the bag
(b) increase the force on the shoulders
(c) spread the force over a larger area, reducing pressure
(d) decrease the area of contact
9. A sharp knife cuts easily because it:
(a) applies less force
(b) concentrates the force on a very small area, giving high pressure
(c) has a larger area of contact
(d) is lighter than a blunt knife
10. People place a cloth pad on their heads while carrying loads in order to:
(a) increase the contact area and reduce pressure
(b) increase the pressure
(c) reduce the weight of the load
(d) increase the force
11. Needles and syringes have sharp, pointed ends so that they:
(a) reduce the force
(b) increase the area
(c) exert high pressure over a small area
(d) decrease the pressure
12. A force of 60 N acts over an area of 2 m². The pressure is:
(a) 120 Pa (b) 62 Pa (c) 58 Pa (d) 30 Pa
13. The area of a tub base measuring 15 cm × 20 cm, in m², is:
(a) 0.3 m² (b) 0.03 m² (c) 0.003 m² (d) 3 m²
14. A tub weighing 60 N has a base area of 0.03 m². The pressure it exerts on the table is:
(a) 200 Pa (b) 2000 Pa (c) 20 Pa (d) 1800 Pa
15. A force of 100 N acts on an area of 0.5 m². The pressure is:
(a) 50 Pa (b) 200 Pa (c) 500 Pa (d) 20 Pa

16. Liquids exert pressure:
(a) only downwards (b) only on the walls
(c) in all directions (d) only upwards
17. The pressure of a liquid at a point depends on:
(a) shape of the container
(b) total amount of liquid
(c) width of the container
(d) height (depth) of the liquid above that point
18. In Activity 1, when the water level in a broad and a narrow pipe is the same, the balloons at the bottom:
(a) bulge more in the broad pipe (b) bulge more in the narrow pipe
(c) bulge equally (d) do not bulge
19. In Activity 1, when more water is added to one pipe, its balloon:
(a) bulges less (b) bulges more
(c) remains unchanged (d) shrinks
20. A tall, narrow column of water exerts the same pressure as a short, wide container if:
(a) their heights are equal (b) their weights are equal
(c) their volumes are equal (d) their widths are equal
21. In Activity 2, water flows out of holes made at the same height of a bottle:
(a) only from the top hole
(b) with different forces
(c) with equal force and reaches the same distance
(d) does not flow
22. Water leaks forcefully from a damaged water pipe because:
(a) the pipe is hot
(b) the water inside pushes against the pipe walls
(c) air outside pushes the water
(d) gravity pulls it sideways
23. The base of a dam is made much broader than the top because:
(a) it stores more water
(b) it is cheaper to build
(c) it must withstand higher water pressure at greater depths
(d) it reduces the flow of water
24. Which part of a dam experiences the strongest sideways pressure?
(a) Top (b) Middle
(c) Bottom (d) All parts equally
25. Water tanks are placed at elevated positions because:
(a) it saves space
(b) the height creates pressure that forces water through pipes
(c) water stays cleaner
(d) evaporation is reduced
26. Submarines have specially reinforced bodies to:
(a) increase their speed
(b) withstand the high pressure of water around them
(c) float easily
(d) reduce their weight
27. Gases exert pressure on:
(a) only the walls of a closed container (b) only the ground
(c) every surface they touch (d) only upward surfaces

28. In Activity 3, it is harder to lift the paper plate when it is covered by:
- (a) a folded sheet of chart paper
 - (b) an unfolded (large) sheet of chart paper
 - (c) no sheet
 - (d) a smaller sheet
29. A balloon expands in all directions when inflated because the air inside:
- (a) presses outward equally on all sides
 - (b) presses only upward
 - (c) is hot
 - (d) is heavier than outside air
30. The layer of air surrounding the Earth is called the:
- (a) hydrosphere
 - (b) atmosphere
 - (c) lithosphere
 - (d) biosphere
31. A rubber sucker sticks firmly to a smooth surface because:
- (a) of glue on it
 - (b) of magnetism
 - (c) the higher outside atmospheric pressure pushes it against the surface
 - (d) of gravity only
32. The force exerted by the air column above an area of $15\text{ cm} \times 15\text{ cm}$ is about:
- (a) 225 N
 - (b) 22.5 N
 - (c) 22500 N
 - (d) 2250 N
33. A force of 2250 N is roughly equal to the weight of a mass of:
- (a) 22.5 kg
 - (b) 225 kg
 - (c) 2250 kg
 - (d) 25 kg
34. We are not crushed by atmospheric pressure because:
- (a) atmospheric pressure is very weak
 - (b) our body fluids and gases exert an equal pressure outwards
 - (c) air pressure acts only upwards
 - (d) gravity cancels it
35. 1 hPa is equal to:
- (a) 1 Pa
 - (b) 10 Pa
 - (c) 100 Pa
 - (d) 1000 Pa
36. Atmospheric pressure is usually expressed in:
- (a) newtons
 - (b) millibars or hectopascals
 - (c) joules
 - (d) metres per second
37. Warm air is:
- (a) heavier and sinks, creating high pressure
 - (b) denser than cool air
 - (c) lighter and rises, creating low pressure
 - (d) unaffected by heating
38. Cool air is heavier and sinks, creating:
- (a) low pressure
 - (b) high pressure
 - (c) no pressure
 - (d) zero pressure
39. Wind is:
- (a) moving water
 - (b) moving air caused by differences in air pressure
 - (c) still air
 - (d) heat from the Sun
40. Air always moves from a region of:
- (a) low pressure to high pressure
 - (b) high pressure to low pressure
 - (c) hot to cold only
 - (d) east to west
41. The greater the difference in air pressure between two places:
- (a) the slower the air moves
 - (b) the wind stops
 - (c) the faster the air moves
 - (d) the cooler the air becomes

42. In Activity 5, after the inflated balloon is released, it:
- (a) becomes smaller while the other balloon grows
 - (b) bursts
 - (c) becomes bigger
 - (d) remains unchanged
43. A sea breeze blows:
- (a) from sea to land during the day
 - (b) from land to sea during the day
 - (c) from land to sea at night
 - (d) from sea to land at night
44. A land breeze blows at night because:
- (a) the land is warmer than the sea
 - (b) air over the warm sea rises, creating low pressure over the sea
 - (c) pressure over the land is lower
 - (d) Earth's rotation pushes the wind
45. In Activity 6, the two hanging balloons move towards each other when air is blown between them because:
- (a) pressure between them increases
 - (b) a low-pressure zone forms between them
 - (c) they attract magnetically
 - (d) gravity pulls them
46. High-speed winds are accompanied by:
- (a) higher air pressure
 - (b) lower air pressure
 - (c) no change in pressure
 - (d) zero pressure
47. During a storm, roofs may be blown away because there is:
- (a) high pressure above the roof and low pressure inside
 - (b) low pressure above the roof and high pressure inside
 - (c) equal pressure on both sides
 - (d) heavy rain only
48. Doors and windows are kept open during a storm to:
- (a) reduce the pressure difference between inside and outside
 - (b) let lightning pass
 - (c) increase the pressure inside
 - (d) make the house lighter
49. Storms are especially common in:
- (a) polar regions
 - (b) hot, humid, tropical regions such as India
 - (c) deserts only
 - (d) cold mountains
50. As rising warm, moist air cools at higher altitude, the water vapour:
- (a) evaporates
 - (b) condenses into droplets that form clouds
 - (c) disappears
 - (d) turns to wind
51. In a storm cloud, the lighter ice particles become:
- (a) negatively charged and stay at the bottom
 - (b) positively charged and move to the top
 - (c) neutral
 - (d) negatively charged and move to the top
52. In a storm cloud, the heavier water droplets become:
- (a) positively charged and move to the top
 - (b) negatively charged and stay near the bottom
 - (c) positively charged and stay near the bottom
 - (d) neutral

53. Under normal conditions, air acts as:
 (a) a conductor (b) an insulator (c) a magnet (d) a battery
54. Thunder is produced due to:
 (a) friction between clouds
 (b) rapid expansion of air heated by lightning
 (c) falling rain
 (d) speed of the wind
55. A storm accompanied by lightning and thunder is called a:
 (a) cyclone (b) thunderstorm (c) sea breeze (d) land breeze
56. The bottom end of a lightning conductor is connected to:
 (a) the roof
 (b) a metal plate buried deep in the ground
 (c) a cloud
 (d) an electric wire
57. During lightning, if you are outdoors, the safest place is:
 (a) under a tall tree (b) near a pond
 (c) inside a car or bus (d) in an open field with an umbrella
58. Which should be avoided during lightning?
 (a) Staying indoors (b) Crouching in a low spot
 (c) Using an umbrella with a metal rod (d) Sitting inside a bus
59. Kalboishakhi, a local thunderstorm, occurs in:
 (a) Kerala (b) West Bengal, Bihar and Jharkhand
 (c) Assam (d) Karnataka
60. Bordoisila is a local thunderstorm of:
 (a) Assam (b) Kerala (c) Tamil Nadu (d) Bihar
61. Mango showers, which help mangoes ripen, occur in:
 (a) Assam and Bihar (b) Kerala, Karnataka and Tamil Nadu
 (c) West Bengal and Jharkhand (d) Punjab and Haryana
62. A cyclone forms over:
 (a) cold mountains (b) warm ocean waters
 (c) deserts (d) polar ice
63. Hurricanes and typhoons are:
 (a) names of cyclones in different parts of the world
 (b) types of thunderstorms
 (c) kinds of sea breeze
 (d) types of lightning
64. Heat released during condensation in a cyclone:
 (a) cools the air
 (b) warms the rising air even more, creating an even lower pressure
 (c) has no effect
 (d) stops the cyclone
65. The Earth's rotation causes the moving air in a cyclone to:
 (a) spin (b) stop (c) cool down (d) condense
66. The eye of a cyclone is:
 (a) the region of strongest winds (b) the region of lowest pressure
 (c) usually calm and clear (d) both (b) and (c)
67. The strongest winds and heaviest rainfall in a cyclone occur in the:
 (a) eye (b) eyewall (c) outer sky (d) ocean floor

68. A cyclone weakens over land because:
- land is cold
 - it no longer receives warm, moist air from the ocean
 - land has many trees
 - land pressure is high
69. The wall of water pushed towards the shore by a cyclone may reach a height of:
- 30–120 m
 - 3–12 m
 - 0.3–1.2 m
 - 300 m
70. Seawater flooding by a cyclone makes farmland soil:
- more fertile
 - less fertile due to salt
 - unchanged
 - richer in minerals
71. Weather warnings about cyclones in India are issued by the:
- NASA
 - India Meteorological Department (IMD)
 - WHO
 - UNESCO
72. Which item is NOT part of an emergency kit for cyclone-prone areas?
- Drinking water
 - Medicines
 - Torch
 - Heavy furniture

B. Fill in the Blanks

- Pressure = Force / _____.
- One pascal is one _____ per square metre.
- Pressure _____ when force increases and the area remains the same.
- Backpacks have _____ straps to spread the force over a larger area.
- Liquids exert pressure on the bottom and on the _____ of their containers.
- The pressure of a liquid _____ with increasing depth.
- Liquid pressure does not depend on the _____ of the container.
- The base of a dam is broader because water pressure is _____ at greater depths.
- Water tanks are placed at _____ positions to supply water to upper floors.
- Air exerts pressure on _____ the surfaces it touches.
- The layer of air surrounding the Earth is called the _____.
- Atmospheric pressure acts in _____ directions.
- 1 mb = 1 hPa = _____ Pa.
- Wind is moving _____.
- Warm air rises and creates a _____ pressure area.
- Cool air sinks and creates a _____ pressure area.
- A sea breeze blows during the _____.
- A land breeze blows at _____.
- High-speed air creates a region of _____ pressure.
- The rubbing of ice particles and water droplets in a cloud produces _____ electricity.
- In a storm cloud, lighter ice particles move to the _____ of the cloud.
- The loud sound heard after lightning is called _____.
- A storm with lightning and thunder is called a _____.
- The bottom end of a lightning conductor is connected to a _____ buried deep in the ground.
- A cyclone forms over warm _____ waters.
- The centre of a cyclone is called the _____.
- The area around the eye with the strongest winds is called the _____.

28. Cyclones weaken when they move over _____.
29. Bordoisila is a local thunderstorm of _____.
30. Cyclone warnings in India are issued by the _____.

C. State whether True (T) or False (F)

1. Pressure and force are the same thing.
2. The SI unit of pressure is pascal.
3. The same force acting on a smaller area produces a greater pressure.
4. Pressure is directly proportional to area.
5. A blunt knife cuts more easily than a sharp knife.
6. Liquids exert pressure only on the bottom of a container.
7. Liquid pressure depends on the total amount of liquid in the container.
8. The pressure of a liquid increases with depth.
9. Jets of water from holes at the same height of a bottle emerge with similar force.
10. The base of a dam is made broader than its top.
11. The higher the water tank, the more pressure it creates.
12. Gases do not exert pressure.
13. Air pressure acts in all directions.
14. The atmosphere has weight.
15. We are not crushed by atmospheric pressure because our body exerts an equal pressure outwards.
16. Warm air is heavier than cool air.
17. Air moves from a low-pressure region to a high-pressure region.
18. A sea breeze blows from the sea to the land during the day.
19. A land breeze blows from the land to the sea at night.
20. High-speed winds create regions of high pressure.
21. Keeping doors and windows open during a storm helps reduce the pressure difference.
22. Storms are especially common in hot, humid, tropical regions.
23. Heavier water droplets in a storm cloud become positively charged and move to the top.
24. Air normally acts as an insulator.
25. Thunder is produced by the rapid expansion of air heated by lightning.
26. It is safe to stand under a tall tree during lightning.
27. A car or bus with a metal body is safer than open ground during lightning.
28. Cyclones form over cold land.
29. The eye of a cyclone has the strongest winds.
30. A cyclone becomes stronger after it reaches land.

D. Match the Following

Set I

Column A

1. Kalboishakhi
2. Bordoisila
3. Mango showers
Nadu
4. Local thunderstorms in Karnataka
5. IMD

Column B

- a. West Bengal, Bihar and Jharkhand
- b. Assam
- c. Help mangoes ripen in Kerala, Karnataka and Tamil Nadu
- d. Support the growth of coffee plants
- undefined. Issues weather reports and warnings

Set II

Column A

1. Pressure
2. Pascal
3. Atmospheric pressure
4. Wind
5. Thunderstorm
the Earth

Column B

- a. SI unit of pressure
- b. Moving air
- c. Force acting per unit area
- d. Storm with lightning and thunder
- undefined. Pressure exerted by the layer of air around the Earth

Set III

Column A

1. Warm air
2. Cool air
3. Fast-moving air
4. Heavier water droplets in a storm cloud
5. Lighter ice particles in a storm cloud

Column B

- a. Negative charge near the bottom of the cloud
- b. Region of low pressure
- c. Positive charge at the top of the cloud
- d. Rises and creates low pressure
- undefined. Sinks and creates high pressure

Set IV

Column A

1. Activity 1 (pipes and balloons)
2. Activity 2 (bottle with holes)
3. Activity 3 (paper plate and chart paper)
4. Activity 4 (rubber sucker)
5. Activity 6 (two hanging balloons)
the container

Column B

- a. Effect of atmospheric pressure
- b. Liquid pressure depends on height of liquid column
- c. Total force due to air pressure increases with area
- d. High-speed air creates low pressure
- undefined. Liquid exerts pressure on the walls of the container

E. Assertion–Reason Questions

Options: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true, but R is false. (d) A is false, but R is true.

1. Assertion (A): Needles pierce the skin easily.
Reason (R): The tip of a needle has a very small area, so it exerts high pressure.
2. Assertion (A): The base of a dam is broader than its top.
Reason (R): The pressure of water increases with depth.
3. Assertion (A): Wide straps make a backpack more comfortable.
Reason (R): Wide straps increase the force acting on the shoulders.
4. Assertion (A): Liquid pressure depends on the total amount of liquid in a container.
Reason (R): Liquid pressure depends on the height of the liquid column.
5. Assertion (A): We are not crushed by atmospheric pressure.
Reason (R): Our body exerts an equal pressure outwards.
6. Assertion (A): Roofs may be blown off during a storm.
Reason (R): High-speed winds create low pressure above the roof.

7. Assertion (A): A lightning conductor is fixed on tall buildings.
Reason (R): Air is normally an insulator.
8. Assertion (A): A sea breeze blows during the day.
Reason (R): During the day, the land cools faster than the sea.
9. Assertion (A): The eye of a cyclone has the strongest winds.
Reason (R): The eyewall experiences the strongest winds and heaviest rainfall.
10. Assertion (A): Cyclones weaken after reaching land.
Reason (R): Land does not supply warm, moist air to the cyclone.

F. Picture / Diagram-based Objective Questions

1. A pencil is pressed between the thumb and a finger. The finger touching the pointed end feels:

(a) more pressure	(b) less pressure
(c) same pressure as the blunt end	(d) no pressure
2. A bottle has three holes at different heights. The jet of water travels farthest from the:

(a) top hole	(b) middle hole
(c) bottom hole	(d) all travel equally
3. In the sea breeze diagram (land warmer, sea cooler), cool air moves:

(a) from land to sea	(b) from sea to land	(c) only upward	(d) nowhere
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4. In the lightning diagram, the bottom of the cloud is negative; the ground and nearby objects become:

(a) positively charged	(b) negatively charged
(c) neutral	(d) alternately charged
5. In the cyclone diagram, the low-pressure region labelled 'Eye' is located:

(a) at the outer edge	(b) at the centre
(c) outside the cyclone	(d) only at the top
6. In Activity 3, the paper plate is easiest to lift when it is covered by:

(a) the folded sheet	(b) the unfolded sheet
(c) both sheets	(d) none of these

G. Numerical-based Objective Questions

1. A force of 500 N acts on an area of 5 m². The pressure is:

(a) 100 Pa	(b) 2500 Pa	(c) 505 Pa	(d) 10 Pa
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2. The pressure is 300 Pa and the area is 4 m². The force is:

(a) 75 N	(b) 1200 N	(c) 304 N	(d) 120 N
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3. A force of 200 N produces a pressure of 50 Pa. The area is:

(a) 10000 m ²	(b) 0.25 m ²	(c) 4 m ²	(d) 150 m ²
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4. The area of a rectangle 40 cm × 50 cm in m² is:

(a) 2 m ²	(b) 0.02 m ²	(c) 0.2 m ²	(d) 20 m ²
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5. A brick weighing 30 N rests on a face of 10 cm × 20 cm. The pressure is:

(a) 150 Pa	(b) 1500 Pa	(c) 15 Pa	(d) 600 Pa
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6. 3000 Pa expressed in hPa is:

(a) 300 hPa	(b) 3 hPa	(c) 30 hPa	(d) 3000 hPa
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7. An atmospheric pressure of 1013 hPa in pascal is:

(a) 1013 Pa	(b) 10130 Pa	(c) 101300 Pa	(d) 1013000 Pa
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8. The same force acts on two surfaces of area 2 m² and 4 m². The ratio of pressures (smaller area : larger area) is:

(a) 1 : 2	(b) 2 : 1	(c) 1 : 1	(d) 4 : 1
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ANSWER KEY

A. Multiple Choice Questions

1-b, 2-d, 3-c, 4-d, 5-b, 6-a, 7-b, 8-c, 9-b, 10-a, 11-c, 12-d, 13-b, 14-b, 15-b, 16-c, 17-d, 18-c, 19-b, 20-a, 21-c, 22-b, 23-c, 24-c, 25-b, 26-b, 27-c, 28-b, 29-a, 30-b, 31-c, 32-d, 33-b, 34-b, 35-c, 36-b, 37-c, 38-b, 39-b, 40-b, 41-c, 42-a, 43-a, 44-b, 45-b, 46-b, 47-b, 48-a, 49-b, 50-b, 51-b, 52-b, 53-b, 54-b, 55-b, 56-b, 57-c, 58-c, 59-b, 60-a, 61-b, 62-b, 63-a, 64-b, 65-a, 66-d, 67-b, 68-b, 69-b, 70-b, 71-b, 72-d

B. Fill in the Blanks

1. Area
2. newton
3. increases
4. wide (broad)
5. walls
6. increases
7. shape
8. higher (greater)
9. elevated (high)
10. all
11. atmosphere
12. all
13. 100
14. air
15. low
16. high
17. day
18. night
19. low
20. static
21. top
22. thunder
23. thunderstorm
24. metal plate
25. ocean
26. eye
27. eyewall
28. land
29. Assam
30. India Meteorological Department (IMD)

C. True or False

1-F, 2-T, 3-T, 4-F, 5-F, 6-F, 7-F, 8-T, 9-T, 10-T, 11-T, 12-F, 13-T, 14-T, 15-T, 16-F, 17-F, 18-T, 19-T, 20-F, 21-T, 22-T, 23-F, 24-T, 25-T, 26-F, 27-T, 28-F, 29-F, 30-F

D. Match the Following

Set I: 1-b, 2-a, 3-c, 4-d, 5-e, Set II: 1-c, 2-a, 3-e, 4-b, 5-d, Set III: 1-d, 2-e, 3-b, 4-a, 5-c, Set IV: 1-b, 2-e, 3-c, 4-a, 5-d

E. Assertion–Reason

1-a, 2-a, 3-c, 4-d, 5-a, 6-a, 7-b, 8-c, 9-d, 10-a

F. Picture-based

1-a, 2-c, 3-b, 4-a, 5-b, 6-a

G. Numerical-based

1-a, 2-b, 3-c, 4-c, 5-b, 6-c, 7-c, 8-b